Evaluating Membership and Broadcast Protocol Combinations in Gossip-Based Systems

Miguel Mestre*
endi@lazzy.net
MEI, DI, FCT, UNL

Alexandre Santos†
lvoid@stress.in
MEI, DI, FCT, UNL

Nelson‡
marbles@cool.se
MEI, DI, FCT, UNL

ABSTRACT

Gossip-based protocols have emerged as an effective approach to implement scalable and resilient broadcast mechanisms in distributed systems. In such systems, nodes rely on membership protocols to maintain partial views of the network, which directly influence the efficiency and reliability of message dissemination. The performance of gossip-based broadcast depends on the interaction between membership and broadcast strategies, particularly under varying system conditions.

Several approaches can be used to construct and maintain these overlays, each offering different trade-offs in terms of reliability, latency, and communication overhead. In this paper, we implement and experimentally evaluate combinations of membership protocols, namely Cyclon and HyParView, with broadcast protocols such as Flood and Eager-Push Gossip, as well as a custom-designed gossip-based solution. Our results show that different protocol combinations exhibit distinct performance characteristics, highlighting trade-offs between reliability and latency, and demonstrating that the choice of membership protocol plays a key role in determining the effectiveness of broadcast mechanisms.

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1 INTRODUCTION

Modern distributed systems often need to send a message to all nodes participating in the system, an operation known as broadcast. The simplest way to do this is flooding, where every node that receives a message immediately forwards it to all of its neighbors. This guarantees that every node will eventually receive the message, but it comes at a steep cost. Because every node forwards to every neighbor, the total number of messages in the network grows very quickly as the system scales, filling up network links and forcing each node to spend time processing and discarding large numbers of duplicate messages. This extra work does not go unnoticed: the congestion and processing overhead introduced by flooding ultimately manifests as increased end-to-end latency, meaning messages take longer to arrive not because the network is slow, but because it is overloaded.

A more efficient alternative is gossip-based broadcast, where instead of forwarding to every neighbor, each node picks only a small random subset of neighbors to forward to. The size of this subset is controlled by a parameter called the fanout. At first glance it might seem that reducing the number of forwarding targets would significantly hurt reliability, but in practice, if the fanout is well calibrated, the message still reaches the vast majority of nodes in the system. The core argument is therefore that the overhead flooding introduces does not meaningfully improve reliability over a well tuned gossip protocol, it only adds congestion and latency.

For gossip to work well, nodes need a way to know which other nodes exist in the system and to maintain a set of neighbors to gossip with. This is the role of an overlay network protocol. In this work we consider two such protocols: Cyclon and HyParView. Cyclon maintains a single list of neighbors and periodically shuffles it with other nodes to keep the view random and fresh. It is simple and works well in stable conditions. HyParView takes a different approach and maintains two lists: an active view of neighbors that are currently in use, and a passive view of backup neighbors that are kept in reserve. When an active neighbor leaves or stops responding, HyParView can immediately replace it with a node from the passive view without waiting for the next maintenance cycle. This makes HyParView more resilient when nodes frequently join and leave the network, a condition known as churn. In scenarios with little churn however, the simpler design of Cyclon is expected to introduce slightly less overhead and therefore slightly lower latency.

On top of these overlay protocols we implement a hybrid gossip broadcast protocol. Rather than always forwarding the full message to neighbors, our protocol splits neighbors into two groups per message. A small subset called eager targets receives the full message right away. Another subset called lazy targets receives only a small announcement indicating that the sender has a message available. If a lazy target has not yet seen that message, it can request the full content, effectively pulling it on demand. This reduces unnecessary full message transmissions when nodes have already received the message through another path. Additionally, our protocol monitors

*Student number 66666. Endi was quite lazy and did not did much on this.
†Student number 77777. Void did stress a lot about this project, he made lots of phone calls and pushed hard for people to work.
‡Student number 11111. Marbles did not understood the assignment but he did use an LLM to make a song related with the project.

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how many duplicate messages it is receiving and uses this information to dynamically adjust the eager fanout over time, growing it when too few nodes seem to be receiving messages and shrinking it when too many duplicates are observed. We expect this adaptive behavior to reduce message overhead compared to a static eager push approach, though the dynamic adjustment of the fanout may occasionally cause short periods of reduced delivery reliability.

In this paper we experimentally study these three aspects: whether flooding truly introduces more latency than gossip for similar reliability levels, whether HyParView outperforms Cyclon under high churn while Cyclon holds an edge in stable networks, and whether our hybrid adaptive protocol improves on static gossip in terms of overhead while remaining competitive in reliability. The remainder of this paper is organized as follows. Section **??** discusses related work on overlay networks and gossip protocols. Section **??** describes our implementation in detail. Section **??** presents and discusses our experimental results. Finally, Section **??** concludes the paper.

2 TEMPLATE OVERVIEW

As noted in the introduction, the "`acmart`" document class can be used to prepare many different kinds of documentation — a double-blind initial submission of a full-length technical paper, a two-page SIGGRAPH Emerging Technologies abstract, a "camera-ready" journal article, a SIGCHI Extended Abstract, and more — all by selecting the appropriate *template style* and *template parameters*.

This document will explain the major features of the document class. For further information, the *LaTeX User's Guide* is available from https://www.acm.org/publications/proceedings-template.

2.1 Template Styles

The primary parameter given to the "`acmart`" document class is the *template style* which corresponds to the kind of publication or SIG publishing the work. This parameter is enclosed in square brackets and is a part of the `documentclass` command:

```
\documentclass[STYLE]{acmart}
```

Journals use one of three template styles. All but three ACM journals use the `acmsmall` template style:

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2.2 Template Parameters

In addition to specifying the *template style* to be used in formatting your work, there are a number of *template parameters* which modify some part of the applied template style. A complete list of these parameters can be found in the *LaTeX User's Guide.*

Frequently-used parameters, or combinations of parameters, include:

- `anonymous,review`: Suitable for a "double-blind" conference submission. Anonymizes the work and includes line numbers. Use with the `\acmSubmissionID` command to print the submission's unique ID on each page of the work.
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This document uses the following string as the first command in the source file:

```
\documentclass[sigconf]{acmart}
```

3 MODIFICATIONS

Modifying the template — including but not limited to: adjusting margins, typeface sizes, line spacing, paragraph and list definitions, and the use of the `\vspace` command to manually adjust the vertical spacing between elements of your work — is not allowed.

Your document will be returned to you for revision if modifications are discovered.

4 TYPEFACES

The "`acmart`" document class requires the use of the "Libertine" typeface family. Your TeX installation should include this set of packages. Please do not substitute other typefaces. The "`lmodern`" and "`ltimes`" packages should not be used, as they will override the built-in typeface families.

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If your title is lengthy, you must define a short version to be used in the page headers, to prevent overlapping text. The `title` command has a "short title" parameter:

```
\title[short title]{full title}
```

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Each author must be defined separately for accurate metadata identification. Multiple authors may share one affiliation. Authors' names should not be abbreviated; use full first names wherever possible. Include authors' e-mail addresses whenever possible.

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```
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\email{dave,judy,steve@university.edu}
\email{firstname.lastname@phillips.org}
```

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```
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```

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9 SECTIONING COMMANDS

Your work should use standard LaTeX sectioning commands: `section`, `subsection`, `subsubsection`, and `paragraph`. They should be numbered; do not remove the numbering from the commands.

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is **not allowed.**

10 TABLES

The "`acmart`" document class includes the "`booktabs`" package — https://ctan.org/pkg/booktabs — for preparing high-quality tables.

Table 1: Frequency of Special Characters

Non-English or Math	Frequency	Comments
Ø	1 in 1,000	For Swedish names
π	1 in 5	Common in math
$	4 in 5	Used in business
Ψ_1^2	1 in 40,000	Unexplained usage

Table captions are placed *above* the table.

Because tables cannot be split across pages, the best placement for them is typically the top of the page nearest their initial cite. To ensure this proper "floating" placement of tables, use the environment **table** to enclose the table's contents and the table caption. The contents of the table itself must go in the **tabular** environment, to be aligned properly in rows and columns, with the desired horizontal and vertical rules. Again, detailed instructions on **tabular** material are found in the *LaTeX User's Guide*.

Immediately following this sentence is the point at which Table 1 is included in the input file; compare the placement of the table here with the table in the printed output of this document.

To set a wider table, which takes up the whole width of the page's live area, use the environment **table*** to enclose the table's contents and the table caption. As with a single-column table, this wide table will "float" to a location deemed more desirable. Immediately following this sentence is the point at which Table 2 is included in the input file; again, it is instructive to compare the placement of the table here with the table in the printed output of this document.

Always use midrule to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognise table headers and support their users in navigating tables more easily.

11 MATH EQUATIONS

You may want to display math equations in three distinct styles: inline, numbered or non-numbered display. Each of the three are discussed in the next sections.

11.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the **math** environment, which can be invoked with the usual `\begin . . . \end` construction or with the short form `$ . . . $`. You can use any of the symbols and structures, from α to ω, available in LaTeX [24]; this section will simply show a few examples of in-text equations in context. Notice how this equation: $\lim_{n\rightarrow \infty}x=0$, set here in in-line math style, looks slightly different when set in display style. (See next section).

11.2 Display Equations

A numbered display equation—one set off by vertical space from the text and centered horizontally—is produced by the **equation** environment. An unnumbered display equation is produced by the **displaymath** environment.

Again, in either environment, you can use any of the symbols and structures available in LaTeX; this section will just give a couple of examples of display equations in context. First, consider the

Table 2: Some Typical Commands

Command	A Number	Comments
`\author`	100	Author
`\table`	300	For tables
`\table*`	400	For wider tables

equation, shown as an inline equation above:

$$\lim_{n\rightarrow \infty}x=0 \tag{1}$$

Notice how it is formatted somewhat differently in the **displaymath** environment. Now, we'll enter an unnumbered equation:

$$\sum_{i=0}^{\infty} x + 1$$

and follow it with another numbered equation:

$$\sum_{i=0}^{\infty}x_i=\int_{0}^{\pi+2} f \tag{2}$$

just to demonstrate LaTeX's able handling of numbering.

12 FIGURES

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Figure 1: 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (https://goo.gl/VLCRBB).

Your figures should contain a caption which describes the figure to the reader.

Figure captions are placed *below* the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what's in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). **Figure descriptions should not repeat the figure caption – their purpose is to capture important information that is not already provided in the caption or the main text of the paper.** For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see https://www.acm.org/publications/taps/describing-figures/.

13 CITATIONS AND BIBLIOGRAPHIES

The use of BiTeX for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete — use full first names ("Donald E. Knuth") not initials ("D. E. Knuth") — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two commands, placed just before the `\end{document}` command:

```
\bibliographystyle{ACM-Reference-Format}
\bibliography{bibfile}
```

where "`bibfile`" is the name, without the "`.bib`" suffix, of the BiTeX file.

Citations and references are numbered by default. A small number of ACM publications have citations and references formatted in the "author year" style; for these exceptions, please include this command in the **preamble** (before the command "`\begin{document}`") of your LaTeX source:

```
\citestyle{acmauthoryear}
```

Some examples. A paginated journal article [2], an enumerated journal article [10], a reference to an entire issue [9], a monograph (whole book) [23], a monograph/whole book in a series (see 2a in spec. document) [17], a divisible-book such as an anthology or compilation [12] followed by the same example, however we only output the series if the volume number is given [13] (so Editor00a's series should NOT be present since it has no vol. no.), a chapter in a divisible book [35], a chapter in a divisible book in a series [11], a multi-volume work as book [22], a couple of articles in a proceedings (of a conference, symposium, workshop for example) (paginated proceedings article) [3, 15], a proceedings article with all possible elements [34], an example of an enumerated proceedings article [14], an informally published work [16], a couple of preprints

[6, 7], a doctoral dissertation [8], a master's thesis: [4], an online document / world wide web resource [1, 28, 36], a video game (Case 1) [27] and (Case 2) [26] and [25] and (Case 3) a patent [33], work accepted for publication [30], 'YYYYb'-test for prolific author [31] and [32]. Other cites might contain 'duplicate' DOI and URLs (some SIAM articles) [21]. Boris / Barbara Beeton: multi-volume works as books [19] and [18]. A couple of citations with DOIs: [20, 21]. Online citations: [36–38]. Artifacts: [29] and [5].

14 ACKNOWLEDGMENTS

Identification of funding sources and other support, and thanks to individuals and groups that assisted in the research and the preparation of the work should be included in an acknowledgment section, which is placed just before the reference section in your document.

This section has a special environment:

```
\begin{acks}
...
\end{acks}
```

so that the information contained therein can be more easily collected during the article metadata extraction phase, and to ensure consistency in the spelling of the section heading.

Authors should not prepare this section as a numbered or unnumbered `\section`; please use the "`acks`" environment.

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